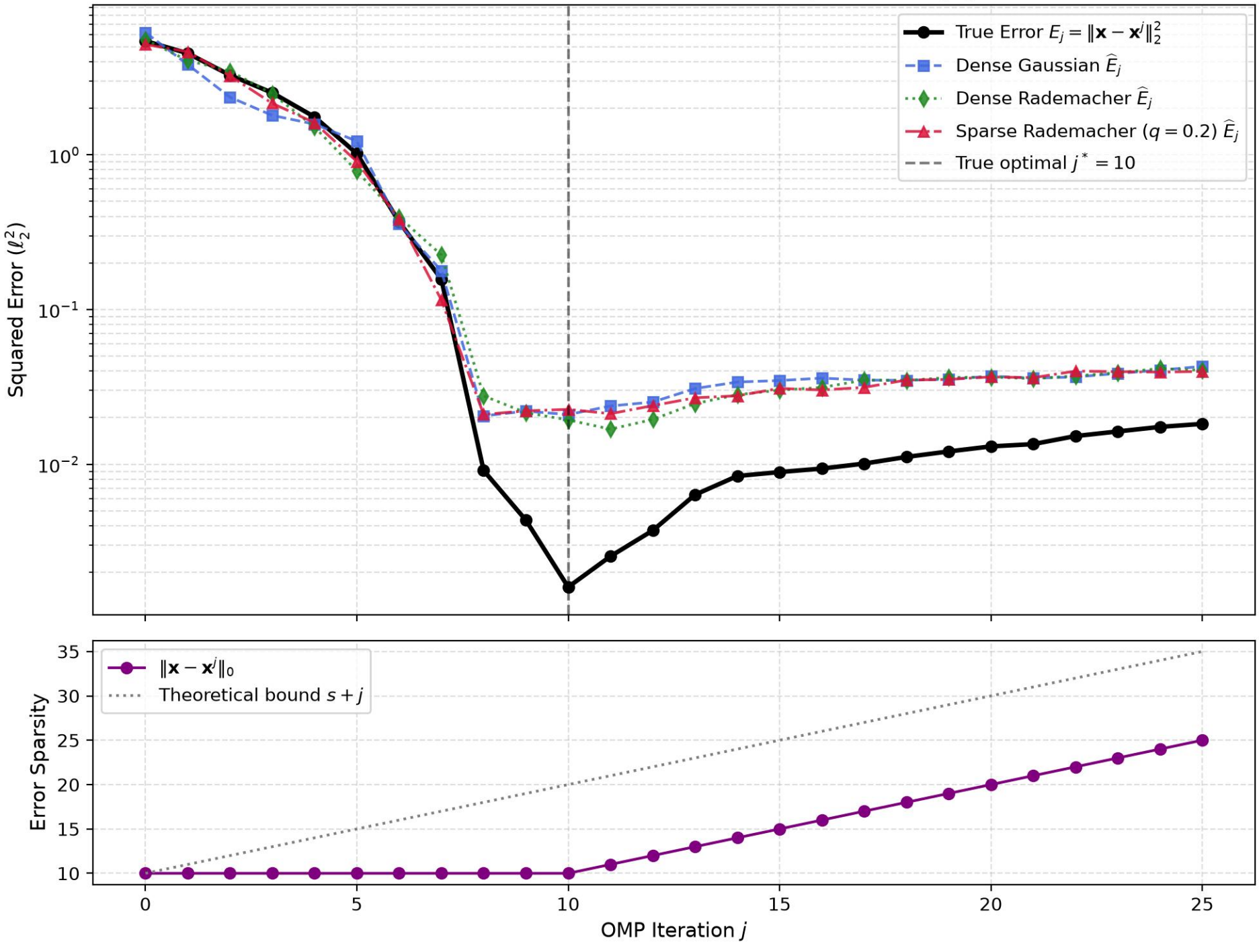

True Error vs Cross-Validation Estimates along OMP Trajectory
(N=512, s=10, m=120, r=40, SNR=25.0 dB, sparse density q=0.2)
True Error $E_j = \|\mathbf{x} - \mathbf{x}^j\|_2^2$
Dense Gaussian $\widehat{E}_j$
Dense Rademacher $\widehat{E}_j$
Sparse Rademacher ($q = 0.2$) $\widehat{E}_j$
True optimal $j^* = 10$
Squared Error (ℓ_2^2)
10^0
10^{-1}
10^{-2}
$\|\mathbf{x} - \mathbf{x}^j\|_0$
Theoretical bound $s + j$
Error Sparsity
35
30
25
20
15
10
0
5
10
15
20
25
OMP Iteration j